EXECUTION PLAN

VC

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

NSX Trust + CM + TNP Repair Plan

Sequenced procedure to repair NSX manager after cold-boot recovery: fix vCenter cert trust (PKIX), re-register Compute Manager (error 40237), and rebuild Transport Node Profile from orphan vDS UUID. All citations Broadcom-documented or lab-verified from prior accomplishments.

NSX 9.0 | PKIX | COMPUTE MANAGER | TNP | CERT TRUST

VCF 9.0.1 NSX 9.0

NSX Trust + CM + TNP Repair Plan

Date: May 15, 2026 **Classification:** Internal — Lab Operations **NSX target:** `192.168.1.71` (single-node cluster, no VIP currently configured/served) **Companion runbooks:** `VCF-Accomplishments-Highlights.md` items 12, 13, 14 (prior NSX cert/TNP work)

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1. Current State (verified 2026-05-15 17:15 UTC)

Item	State	Source
NSX VM	PoweredOn on esxi01.lab.local, uptime ~1h post-cold-boot	PowerCLI
Port 22 (SSH)	UP	TCP probe
Port 443 (HTTPS API/UI)	UP after `start service http`	TCP probe + NSX CLI
`http` service	Running (just started)	`get services`
`manager` service	"Not running or not ready" - stabilizing	NSX CLI error
`controller` service	Stopped — Listen address blank	`get services`
`cm-inventory` service	Stopped — depends on CM	`get services`
`messaging-manager` + `nsx-message-bus`	Stopped	`get services`
NSX → vCenter SSL handshake	**PKIX path validation FAILED — TrustAnchor found but certificate validation failed**	`/var/log/proton/nsxapi.log`
Compute Manager 259b8edb registration	UNREGISTERED, Connection Down	NSX API
Compute Manager error code	**40237 (Service account is not present)** + secondary "Unknown error in processing request" (cert-related)	NSX UI
Transport Nodes (×4) reference vDS	UUID `50 37 13 3f 96 e4 3e 93-1d f0 3b 01 d4 76 6a f8` (`vcenter-cl01-vds01-new`)	NSX API
That vDS in vCenter inventory	**DELETED** — only `vcenter-cl01-vds01` (UUID `50 12 35 19 fa f5 be 0a-71 8d ab e0 b5 e6 b9 cf`) exists	PowerCLI
Active vSphere alert	"vDS proxy switches on the host does not exist in vCenter or does not contain this host"	vCenter UI

Net: NSX is partially operational. Trust + CM + TNP layers are all broken from cascading effects of the May 13 vCenter cert renewal + this morning's NSX recovery work.

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2. Three-Layer Diagnosis

Layer	Symptom	Root cause	Fix
L1 — Cert trust	PKIX path validation fails when NSX manager opens HTTPS to `https://192.168.1.69:443/rest/com/vmware/cis/session`	NSX's java truststore lacks vCenter's current leaf and/or VMCA root (vCenter cert was renewed 2026-05-13)	Update CM registration with current SHA-256 thumbprint OR import VMCA root into NSX truststore
L2 — Compute Manager	Error 40237 "Service account is not present on the compute manager"	NSX's service account on vCenter (e.g., `svc-nsxt-<id>@vsphere.local`) was deleted or invalidated. NSX still has stored creds that vCenter no longer recognizes	Click **"Resolve"** in NSX UI → provide vCenter admin creds → NSX re-creates the service account
L3 — Transport Node Profile	All 4 TNs reference orphan vDS UUID; vSphere shows "proxy switches… vDS does not exist" alarm	NSX TNP `vcenter-cl01-tnp-01` `host_switch_id` points at the old `vcenter-cl01-vds01-new` that was deleted post-recovery in accomplishment #13	Policy API PUT with current vDS UUID + correct uplink mapping + `_revision`

L1 must be fixed BEFORE L2 (L2's Resolve dialog uses HTTPS to vCenter — without trust, Resolve will fail). L2 must be fixed BEFORE L3 (TNP API calls require CM to be operational for inventory sync). So strict order: **L1 → L2 → L3.**

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3. Pre-Execution Gates

All four must pass before Phase 1:

#	Gate	Verify with
G1	NSX port 443 UP	`Test-NetConnection 192.168.1.71 -Port 443`
G2	NSX UI loads `https://192.168.1.71/` in browser	Manual check
G3	Error 101 cleared (no "Component health: Unknown")	UI banner
G4	vCenter SHA-256 thumbprint verified current	PowerShell cert probe (already computed: `3C:4E:9B:ED:CE:B6:EA:5B:15:57:2D:78:0F:85:87:73:80:74:E8:49:68:F5:76:AE:F3:BA:E1:C9:2C:BD:68:5F`)

If G3 fails — wait additional time (per Undocumented Issue #11, NSX needs 10–15 min after service start to stabilize).

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4. Phase 1 — Fix NSX → vCenter Cert Trust (L1)

Step 1.1 — Capture current vCenter cert thumbprint (already done)

```
SHA-256: 3C:4E:9B:ED:CE:B6:EA:5B:15:57:2D:78:0F:85:87:73:80:74:E8:49:68:F5:76:AE:F3:BA:E1:C9:2C:BD:68:5F
```

Step 1.2 — Update CM registration with new thumbprint via NSX UI (preferred)

1. NSX UI → **System** → **Fabric** → **Compute Managers**
2. Click **"Edit"** on the entry for `192.168.1.69`
3. Update **SHA-256 Thumbprint** field with the value above
4. Provide credentials: `administrator@vsphere.local` / `<LAB_PW>`
5. Click **Save**

Step 1.3 — Alternative: thumbprint update via NSX Policy API (if UI fails)

```
curl -k -u 'admin:<LAB_PW>' \
  -X PUT https://192.168.1.71/api/v1/fabric/compute-managers/259b8edb-3b54-496f-bfb4-c85c145cd9bb \
  -H 'Content-Type: application/json' \
  -d '{
    "server": "192.168.1.69",
    "origin_type": "vCenter",
    "credential": {
      "credential_type": "UsernamePasswordLoginCredential",
      "username": "administrator@vsphere.local",
      "password": "<password>",
      "thumbprint":
"3C:4E:9B:ED:CE:B6:EA:5B:15:57:2D:78:0F:85:87:73:80:74:E8:49:68:F5:76:AE:F3:BA:E1:C9:2C:BD:68:5F"
    },
    "_revision": <current_revision_from_GET>
  }'
```

Gate to Phase 2: `tail /var/log/proton/nsxapi.log` no longer shows `PKIX path validation failed`.

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5. Phase 2 — Resolve Compute Manager 40237 Service-Account Error (L2)

Step 2.1 — Click "Resolve" in NSX UI

1. NSX UI → **System** → **Fabric** → **Compute Managers**
2. Click the **"Down"** text in **Connection Status** column (NOT the kebab)
3. Side panel opens with error 40237
4. Click **"RESOLVE"** button
5. Provide credentials: `administrator@vsphere.local` / `<LAB_PW>`
6. NSX recreates service account on vCenter (creates new entry in `vsphere.local` SSO domain, name pattern `svc-nsxt-<id>@vsphere.local`)

Step 2.2 — Verify

Wait 30 sec then re-check Compute Managers list: - Registration Status: **Registered** (green) - Connection Status: **Up** (green) - Last Inventory Update: recent timestamp (within minutes)

Gate to Phase 3: CM shows Registered + Up.

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6. Phase 3 — Repair Transport Node Profile from Orphan vDS (L3)

This is the most complex phase. Source: `VCF-Accomplishments-Highlights.md` item 14 — operator has done this exact procedure before.

Step 3.1 — GET current TNP shape

```
curl -k -u 'admin:<LAB_PW>' \
  https://192.168.1.71/policy/api/v1/infra/host-transport-node-profiles/vcenter-cl01-tnp-01 \
  -o /tmp/tnp-current.json
```

Inspect to capture: - Current `_revision` (must be passed in PUT) - Current `host_switch_profile_id` (may have typo — per accomplishment #14: actual ID was `center-cl01-up-host01`, NO leading "v", despite display_name being `vcenter-cl01-up-host01`) - Current uplink mapping

Step 3.2 — Build corrected PUT payload

Replace `host_switch_id` with current vDS UUID:

```
{
  "host_switch_id": "50 12 35 19 fa f5 be 0a-71 8d ab e0 b5 e6 b9 cf",
  "host_switch_name": "vcenter-cl01-vds01",
  "host_switch_profile_ids": [
    { "key": "UplinkHostSwitchProfile", "value": "center-cl01-up-host01" }
  ],
  "uplinks": [
    { "uplink_name": "uplink-1", "vds_uplink_name": "dvUplink1" },
    { "uplink_name": "uplink-2", "vds_uplink_name": "dvUplink2" }
  ],
  "_revision": <from_GET>
}
```

Three documented gotchas (accomplishment #14):

1. **Uplink names**: PowerCLI default is `dvUplink1` / `dvUplink2`, NOT `uplink1` / `uplink2` → error 9536 if mismatched
2. **Host-switch-profile-id**: actual id may be `center-cl01-up-host01` (no leading "v") despite display_name `vcenter-cl01-up-host01` → error 600 if you trust the display name
3. `_revision` **required**: Missing returns error 500071

Step 3.3 — PUT to update TNP

```
curl -k -u 'admin:<LAB_PW>' \
  -X PUT https://192.168.1.71/policy/api/v1/infra/host-transport-node-profiles/vcenter-cl01-tnp-01 \
```

```
  -H 'Content-Type: application/json' \
  -d @/tmp/tnp-new.json
```

Step 3.4 — Verify TN realization

```
curl -k -u 'admin:<LAB_PW>' \
  https://192.168.1.71/api/v1/transport-nodes | \
  python3 -m json.tool | grep -E '"display_name"|"host_switch_id"'
```

All 4 TNs should now show `host_switch_id` = current vDS UUID.

Step 3.5 — Verify TEP fabric

```
# Inside ESXi01 host:
vmkping -S vxlan -I vmk10 <peer_TEP_ip> -c 3 -s 1572
```

Expected: 0% loss, MTU 1700 OK.

Gate to Phase 4: vCenter alarm "vDS proxy switches... does not exist" cleared.

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7. Phase 4 — Verification

Step 4.1 — Cluster health

```
NSX CLI:
  get cluster status # expect STABLE for control + mgmt
  get services # most services Running
  get managers # single node Connected
```

Step 4.2 — Compute Manager

```
NSX UI → System → Fabric → Compute Managers:
  vCenter: Registered + Up
  Last Inventory Update: within minutes
  Alarms: 0
```

Step 4.3 — Transport Nodes

```
NSX UI → System → Fabric → Nodes → Host Transport Nodes:
  All 4 ESXi hosts: Configuration State = Success
  All 4 ESXi hosts: Node Status = Up
```

Step 4.4 — vCenter alarms

vSphere UI → Monitor → Issues and Alarms: - "vDS proxy switches" alarm cleared - No new NSX-related alarms

Step 4.5 — Skyline Health

vSphere Skyline → no NSX-related red/yellow

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8. Rollback

If Phase 1 (thumbprint update) leaves NSX in a worse state: - Revert to a CM Delete + Re-add (forces fresh registration; loses TN bindings temporarily — Phase 3 fixes them anyway)

If Phase 3 PUT fails repeatedly: - Check exact error code against the three gotchas above - If consistent failures: DELETE the TNP via Policy API, then ADD via NSX UI with correct vDS

If NSX manager crashes mid-procedure: - Another cold boot (Stop-VM -Kill + Start-VM) is a known-safe recovery - Restart sequence: wait for `manager` service, then `start service http`, then `start service controller`

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9. Source Authority

Phase	Source
L1 cert trust update	[BROADCOM-DOC] NSX TechDocs — Edit Compute Manager Registration; `VCF-Accomplishments-Highlights.md` #12
L2 service account 40237	NSX UI error 40237 self-resolve workflow; matches accomplishment #12
L3 TNP rebuild	`VCF-Accomplishments-Highlights.md` #14 (lab-verified pattern); NSX Policy API documentation
Error 101 wait	Undocumented Issues Reference — issue #11 (lab-verified)
NSX boot storm timing	Undocumented Issues Reference — issue #9 (lab-verified)

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10. Operator vs. Automated Tasks

Phase / Step	Who	Why
Pre-execution gates G1–G3	Operator + automated	G1 automated, G2/G3 require UI
1.2 UI thumbprint update	Operator	Web UI workflow
1.3 API thumbprint update (fallback)	Automated	curl + plink
2.1 Click Resolve	Operator	UI workflow
3.1–3.4 TNP via Policy API	Automated (with explicit operator approval before each PUT)	Touching NSX TN config — requires confirmation

Phase / Step	Who	Why
4 Verification	Mostly automated	API calls

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11. Document Information

Field	Value
Plan version	1.0
Date prepared	2026-05-15
Author	Virtual Control LLC
Source events	2026-05-13 vCenter cert renewal; 2026-05-15 NSX cold boot 15:45 UTC
Companion plans	`vCLS-Retreat-MM-Retry-DiskFix-Plan-2026-05-15.md` (completed earlier today)
